AQA

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

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Forename(s) _____

Candidate signature _____

I declare this is my own work.

# Functional Skills Level 1 MATHEMATICS

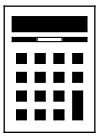
## Paper 2 Calculator

Tuesday 25 February 2020 Afternoon Time allowed: 1 hour 30 minutes

**Materials**

For this paper you must have:

- a calculator
- mathematical instruments.



| For Examiner's Use | |
|---|---|
| Question | Mark |
| 1–8 | |
| 9 | |
| 10 | |
| 11 | |
| 12 | |
| **TOTAL** | |

**Instructions**

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.

**Advice**

In all calculations, show clearly how you work out your answer.

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## Section A

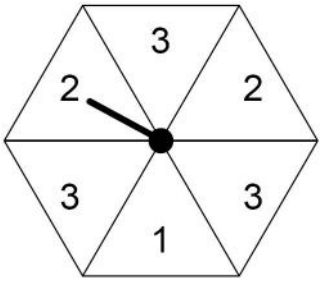
Answer **all** questions in the spaces provided.

**1** Which type of angle is the largest?
Circle your answer.

**[1 mark]**

obtuse reflex acute right

**2** This fair six-sided spinner is spun once.




Draw an arrow on the scale to show the probability of landing on an odd number.

**[1 mark]**




**3** Write in digits one hundred and twenty four thousand six hundred and fifty

**[1 mark]**

Answer _____

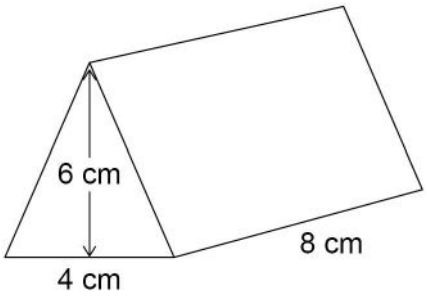
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**4** A triangular prism has width 4 cm, height 6 cm and length 8 cm

**Sketch** a net of the prism.

Label the given dimensions on your net.

**[2 marks]**




**Turn over for the next question**

*Do not write outside the box*

**5** Round 7.8652 to 2 decimal places.

**[1 mark]**

Answer _____

**6** Convert 1200 grams to kilograms.

**[2 marks]**

Answer _____ kilograms

*Do not write outside the box*

**7** How many days are there in October?

**[1 mark]**

Answer _____

**8** Increase 250 by 30%

**[3 marks]**

_____

_____

_____

_____

_____

Answer _____

**12**

**Turn over for Section B**

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## Section B

Answer **all** questions in the spaces provided.

**9** **Dance studio**

Megan owns a dance studio.

**9 (a)** One day, Megan starts teaching her dance classes at 5.30 pm

Megan

- teaches two classes, each lasting 45 minutes
- teaches one class lasting 30 minutes
- has 10-minute breaks between classes
- needs 25 minutes to tidy up and lock up after the last class.

She plans to meet up with friends at a restaurant after her classes.

It takes 12 minutes to walk from the dance studio to the restaurant.

Does she get to the restaurant by 8.30 pm?
You **must** show your working.

**[4 marks]**

*Do not write outside the box*

**9 (b)** Megan is starting a new dance class.

The floor space available for the dancers is a rectangle measuring 13.8 m by 7.1 m

Each dancer needs 4 $m^2$ of floor space.

Work out the maximum number of dancers that can be in the dance class.

**[4 marks]**

Answer _____

**Question 9 continues on the next page**

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**9 (c)** Megan is shopping for a new speaker for her dance studio.

She sees offers in two shops for the speaker.

| G Tone Music Shop<br>Speaker<br>Usual price £129<br>Special offer<br>$\frac{1}{3}$ off | Oma's Music Store<br>Speaker<br>Usual price £112<br>Special offer<br>25% off |
|---|---|

Megan wants to pay the lowest price possible.

Which of these shops should she choose?
You **must** show your working.

**[5 marks]**

Answer _____

13

*Do not write outside the box*

**10 Indoor climbing centre**

**10 (a)** At a climbing centre there must be at least 1 member of staff for every 7 climbers.
There are 6 members of staff for one session.
23 climbers have booked for this session.

Work out how many **more** climbers can book for this session.

**[3 marks]**

Answer _____

**Question 10 continues on the next page**

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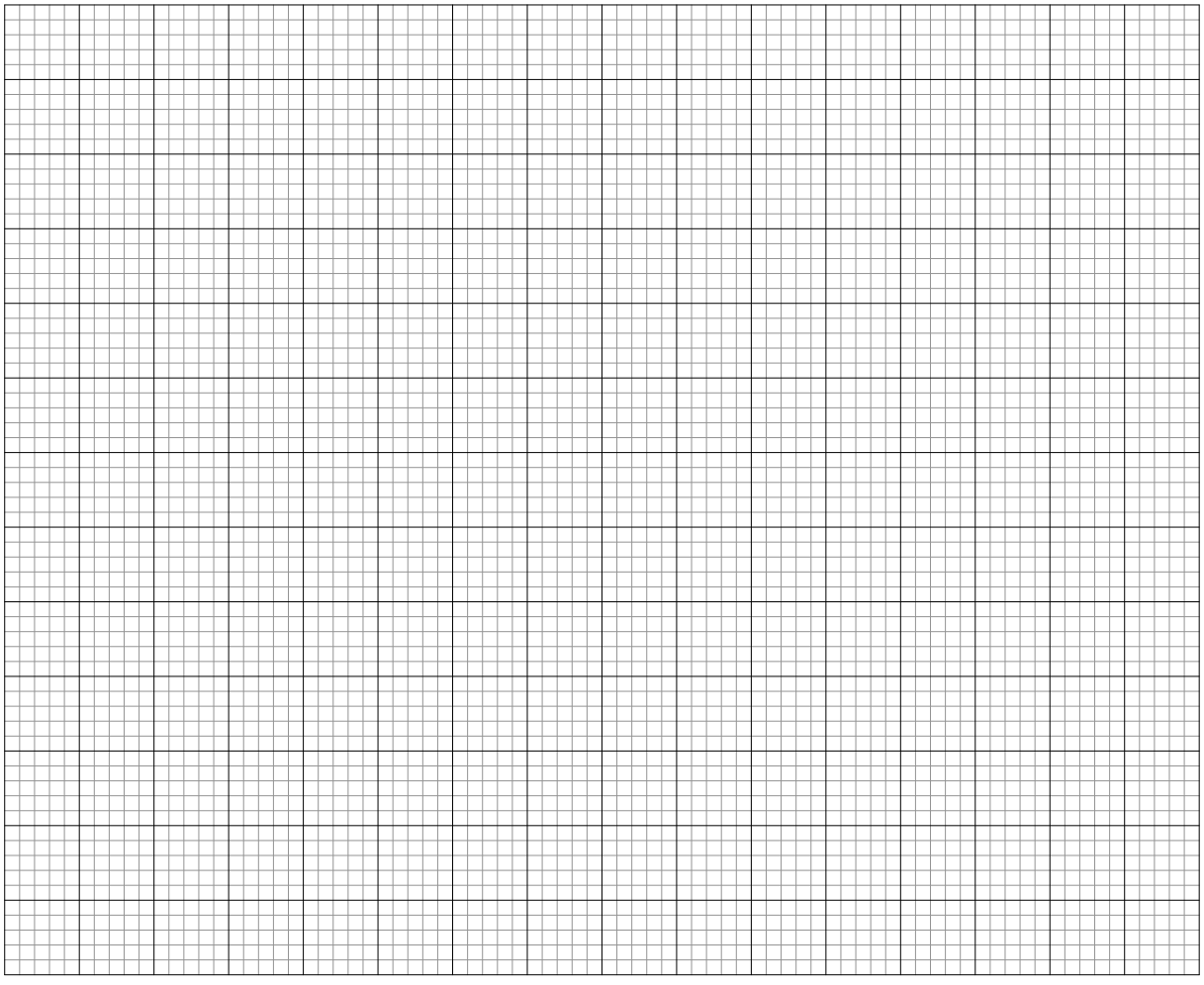
**10 (b)** The table shows the number of climbers that visited the climbing centre on Monday to Friday one week.

| | **Monday** | **Tuesday** | **Wednesday** | **Thursday** | **Friday** |
|---|---|---|---|---|---|
| **Adults** | 18 | 15 | 12 | 21 | 18 |
| **Children** | 23 | 21 | 22 | 23 | 20 |

Show this information on a suitable diagram.

Use the grid below.

**[5 marks]**



*Do not write outside the box*

**10 (c)** The climbing centre organises a competition for climbers.

Each climber has **six** attempts at climbing a 15-metre wall.

To qualify for the final a climber must have a mean time of **less than** 8 seconds.

Here are Oliver's times, in seconds, for his six attempts.

9.22 8.58 7.79 7.23 8.66 8.14

Does Oliver qualify for the final?
You **must** show your working.

**[4 marks]**

12

**Turn over for the next question**

*Do not write outside the box*

**11 Back to school**

**11 (a)** Sandra is shopping for items of school uniform for her children Fred and Beth.
The shopping list below shows the prices of the items Sandra needs to buy.

| **Fred** | **Beth** |
|---|---|
| 2 school jumpers - £17.50 each | 2 packs white shirts - £11.99 per pack |
| 3 grey trousers - £10.75 each | 2 skirts - £9.50 each |
| 1 sports polo shirt - £14 | 1 sports polo shirt - £14.50 |
| 1 pack white shirts - £12.95 per pack | 1 sports skort - £9 |
| School shoes - £45 | 1 rugby shirt - £22.95 |
| | School shoes - £38 |

Fred says,

"The total cost of my uniform is lower than Beth's."

Is Fred correct?
You **must** show your working.

**[5 marks]**

*Do not write outside the box*

**11 (b)** Beth earns pocket money by doing jobs at home after school.

Sandra uses this formula to work out how much pocket money Beth receives **each week**.

| Weekly pocket money = £2.50 × number of jobs done + £5 |
|---|

Beth wants to buy a pair of trainers for £70

She saves her pocket money for 3 weeks.

She did

- 5 jobs in the first week
- 9 jobs in the second week
- 10 jobs in the third week.

Is this enough to buy the trainers?
You **must** show your working.

**[4 marks]**

9

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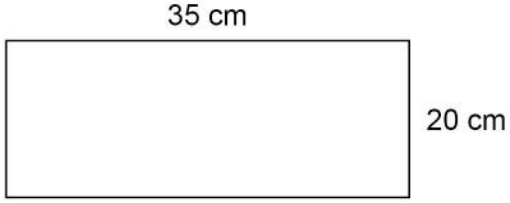
**12 Handbags**

Jon makes handbags to sell online.



Jon wants to make 50 handbags.

**12 (a)** For each handbag Jon needs a rectangle of leather measuring 35 cm by 20 cm




He cuts the rectangles from sheets of leather measuring 120 cm by 100 cm

The grid on the opposite page is a scale drawing of a sheet of leather.

Jon says,

"I need 4 sheets of leather to make 50 handbags."

Is he correct?
You may use the grid to help you.
You **must** show your working.

**[5 marks]**

*Do not write outside the box*

**Scale**: 1 cm represents 10 cm



**Question 12 continues on the next page**

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**12 (b)** Jon fixes a chain to each handbag to make the handle.

Each chain handle is 75 cm long.

He needs enough chain for the 50 handbags.

Jon has 5 rolls of chain.

Each roll of chain is 6 metres long.

How many **more** 6-metre rolls of chain does he need?
You **must** show your working.

**[5 marks]**

Answer _____

*Do not write outside the box*

**12 (c)** The table shows the costs for making the 50 handbags.

| | **Cost for 50 handbags** |
|---|---|
| Leather | £252.50 |
| Chain | £33.50 |
| Fasteners | £119.50 |
| Other costs | £87.00 |

Jon expects to sell all 50 bags.

He wants to make at least £3.50 profit on each bag.

Work out the **minimum** price he should sell each bag for.

**[4 marks]**

Answer £ _____

14

**END OF QUESTIONS**

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**DO NOT WRITE ON THIS PAGE**
**ANSWER IN THE SPACES PROVIDED**

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| Question number | **Additional page, if required.**<br>**Write the question numbers in the left-hand margin.** |
|---|---|
| | |

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